

ASSIGNMENT-2

Identify a Real-world application for both parallel computing and Network systems. Explain how these technologies are used and why they are important in that context.

The real-world Application is: Online Multiplayer Gaming.

Parallel computing use: Online Multiplayer games require real-time processing of vast amounts of data, including player inputs, physics simulations, and rendering graphics. Parallel computing is used on game servers and player devices to handle multiple tasks simultaneously. For instance, game physics, AI behaviors, and environmental effects are often computed in parallel using multi-core processors or CPUs. This improves performance, reduce lag, and delivers smooth gameplay.

Network system use: Network systems enable communication between players and game servers over the internet. They synchronize game state, manage user accounts, and facilitate matchmaking. Technologies like client-server architecture, TCP/UDP protocols, and cloud-based networking platforms ensure reliable and low latency communication between players across the globe.

Importance: These technologies are critical to delivering a seamless and immersive experience. Parallel computing enhances performance and realism, while networked systems make real-time interaction and global multiplayer possible.